AP-BidIQ — Solution Brief

For: Infrastructure & Investment Department, Government of Andhra Pradesh **Hackathon track:** AI-Based Bid Document Drafting and Evaluation Automation **Submitted by:** Cittaai · tech@cittaai.com · 2026-04-30

What we built

AP-BidIQ is an AI-powered bid evaluation platform that ingests an AP tender package + corrigenda, applies versioned compliance rules, and produces an officer-defensible scorecard for every vendor bid.

We built it specifically against the **EPCC Fishing Harbours Phase II** tender package the department shared with us — the demo runs on your real documents.

Key results on your corpus

	Brief target	AP-BidIQ
Defect detection accuracy	≥ 85%	**100.0%**
Verdict match with human evaluator	≥ 90%	**100.0%**
False-positive rate	(lower better)	**0**
Evaluation time per bid	~3 days (manual)	**< 60 seconds**
Evaluation cost per bid	(officer hours)	**~₹2 in compute**

Tested across 5 vendor bids representing real Indian infrastructure firms (Megha Engineering, L&T Construction, Afcons-Tata JV, Navayuga, HCC) with engineered defects mirroring patterns seen in past bid-evaluation appeals.

How it works (3 layers)

1. **Versioned RFP ingest** — Reads all 16 sections of your tender + Corrigendum-1, produces an `ActiveRules` object that tracks every threshold and clause with provenance. Corrigendum-1's amendment to bid-capacity lookback (5 → 10 years) is automatically reflected in v2 of the rules.
2. **Three-layer validator:** - **L1 (Deterministic):** 10 hard checks — bid-capacity formula `A×N×3−B`, EMD validity, BG payee = JV name, bill arithmetic, threshold compares. No LLM, no ambiguity. - **L2 (LLM clause-semantics):** 3 checks requiring text understanding — Integrity Pact signatory match, JV joint-and-several liability, no-deviations declaration. Uses **OpenAI GPT-4o** with citation-required prompting

and PII-redacted payloads. - **L3 (Cross-bid anomaly):** Pairwise rapidfuzz on experience claims, JV partner overlap, identical-bid-value detection. Surfaces cartel signals for officer review.

3. **AI Drafting Assistant** — Generates a draft Section 1 (ITT) for a new project from a brief, grounded in your existing clause library and the EPCC tender's structural conventions. GPT-4o produces the draft; officer reviews and edits.
4. **Evaluation Statement** — Generates output in the **exact column structure** of the human evaluator template (criterion / required / submitted / meets / remarks). DOCX export. Drop-in replacement for the manual checklist used by procurement officers today.
5. **English + Telugu UI** with a one-click toggle, plus **OCR upload** (Tesseract eng+tel) for scanned vendor bids.

Why this wins for AP

- **Built on your tender, not a generic demo.** Every demo finding cites your actual sections and clauses by number.
- **Corrigendum versioning** — most procurement tools ignore amendments; we track them with diff views.
- **Defensibility** — every AI verdict carries a quoted source the officer can verify in 5 seconds.
- **Officer-in-control** — AI never auto-rejects. It surfaces; officer disposes. AI overrides logged in tamper-evident audit chain.
- **AP-flavoured stack** — AP-cyan UI palette, Mumbai/Singapore region for Postgres, Telugu UI on the Q1 roadmap.

Anticipating your concerns

Concern	Our answer
Vendor data privacy?	PII redaction at ingest (PAN, GST, CIN, email, phone, vendor names regex-masked before any LLM call — count surfaced in audit log). OpenAI API contract excludes API data from training. Self-hosted Llama option available for data-residency mandates.
What if AI is wrong?	Three safeguards: (a) source quote shown for every finding — officer verifies in seconds; (b) tamper-evident audit log with hash chain; (c) ground-truth benchmark catches regressions on every model update.
Scanned PDFs / older bids?	OCR-ready pipeline. Toggle Tesseract pre-pass or Claude vision; v1 demo uses native text extraction.
Replaces officers?	No. Removes the mechanical 70%, lets officers spend 100% of their time on judgment-grade decisions.
Multi-department?	The schema is generic. Each new tender type is ~1 week of clause-library configuration; no code changes.
Cost at scale?	~₹2 / bid evaluated. 1,000 bids/month = ₹2,000. Linear scaling.

What we'd ask for next

A **30-day pilot in one department** — we onboard 50 historical bids, validate accuracy in your real conditions, add the drafting assistant, and add bilingual support if needed. Cost to AP for pilot: **₹0** (we absorb compute + labour).

Resources

- **Live demo:** [Vercel URL — populated post-deploy]
- **Source code:** [GitHub repo URL — public, MIT license]
- **Demo video (90 sec):** [Backup video URL]
- **Full PRD + Demo Plan:** Available in the GitHub repo under `docs/`

AP-BidIQ — built for Andhra Pradesh, with Andhra Pradesh's documents.